

## Cover — Opening the meeting

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- **We inserted a flow-matching layer between the frozen features from Stage 1 and the prototype classifier — a small network that moves each feature toward its class prototype before we classify it.**
- **Everything else is unchanged from Stage 1: same encoders, same features, same prototypes, same splits — so any accuracy change is only the flow layer.**
- **Three headline results, up front:** it works — up to +22.3 points. The fancier version (rolled-out) does not beat the simple version. One number predicts, almost perfectly, which cells improve.

### IF HE ASKS

#### Who did what?

State your and Mira's split honestly, same as Stage 1 — whichever notebooks each of you owns.

## Stage 1 refresher — the protocol and the baseline

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- **Stage 1 froze two encoders — ResNet-18 and DINOv2 — and never fine-tuned them. Two classifiers were built on the cached features: a trained linear probe, and an untrained prototype classifier (nearest class-mean, cosine similarity).**
- **Stage 2 changes exactly one thing: it adds the flow layer before classification. Encoders, features, k-shot subsets, seeds, and test splits are all reused unchanged.**
- **The Prototype column is our baseline — it has no flow layer in it yet. It is the number every later result is measured against.**
- **The last column, headroom, is trained-probe minus prototype. Small on DTD, huge on Aircraft. Remember this — it comes back at the very end of the report.**

### IF HE ASKS

#### **What exactly is 'the prototype'?**

No training at all. Average the (normalized) features of each class's training images into one vector per class. To classify a test image, pick whichever class's average vector it is most similar to (cosine similarity).

#### **Is the Prototype column already the flow-matching result?**

No — it is the plain Stage 1 baseline, with no flow layer. The FM numbers start on page 5.

#### **Why no error bar on the full-data rows?**

Full data has no subsampling left to vary — it's one deterministic run. Every K=5 / K=10 row has 3 seeds and does carry an error bar.

#### **Why is headroom near zero or negative on DTD but huge on Aircraft?**

DTD is 47 broad texture classes, where a simple class mean already summarizes the class well. Aircraft is 100 fine-grained variants, where a trained classifier captures a lot that a class mean misses.